Advanced Circuits: Operational Amplifiers and Filters

Problem Set with Solutions

Companion to the Reference Encyclopedia

August 8, 2026

This problem set contains 24 problems spanning the op-amp and filter unit: conceptual questions (C), analytical calculations (A), and Python exercises (P). Full solutions begin in Section 2; make an honest attempt before looking. Problems marked P expect a short script using `numpy`/`scipy`/`matplotlib`.

1 Problems

Op-Amp Fundamentals

Problem 1 (C) — The comparator's secret

A classmate claims: "An op amp is an amplifier, so if I connect a 1 mV audio signal directly between the two inputs of a bare op amp, I'll get a nice amplified copy at the output." Using the fundamental op-amp equation and typical values of A_o and supply voltage, explain quantitatively what will actually happen, and name the one function a feedback-free op amp performs well.

Problem 2 (A) — Linear window

An op amp has $A_o = 5 \times 10^4$ and runs from ± 12 V supplies, saturating at ± 10.8 V. It is wired as a noninverting comparator (V^- grounded). (a) Over what range of input voltages is the output *not* saturated? (b) What differential input produces $V_{\text{out}} = +5$ V? (c) Why is this "linear amplifier" useless in practice?

Problem 3 (C) — Golden rules, fine print

State the two Golden Rules of op-amp analysis. Then give two distinct circuit situations in which Golden Rule 2 ("$V^+ = V^-$") is *invalid*, and explain in one sentence each why the rule breaks down there.

Amplifier Configurations

Problem 4 (A) — Inverting design

Design an inverting amplifier with a voltage gain of -22 and an input impedance of at least 15 kΩ. (a) Give exact resistor values. (b) The nearest standard 5% values are 15 kΩ and 330 kΩ; compute the resulting gain and its percent deviation from -22. (c) What is the circuit's input impedance, and why is it not "nearly infinite" as for the noninverting amplifier?

Problem 5 (A) — Noninverting, ideal and real

A noninverting amplifier uses $R_1 = 2.2$ kΩ (to ground) and $R_F = 47$ kΩ. (a) Compute the ideal gain. (b) Recompute with a finite open-loop gain $A_o = 10^5$ using $G = A_o/(1 + A_o\beta)$, and give the percent error. (c) What value of R_F turns this circuit into a voltage follower, and what is a follower for?

Problem 6 (A) — Summing amplifier as a mixer

Design a summing amplifier with $V_{\text{out}} = -(2V_1 + 0.5V_2)$ using a feedback resistor $R_F = 100$ kΩ. Give both input resistors, and state the output for $V_1 = 1.5$ V, $V_2 = -2$ V.

Problem 7 (A) — Integrator ramp

An op-amp integrator has $R = 100$ kΩ and $C = 100$ nF, powered from ± 12 V (saturation ± 11 V), with the capacitor initially uncharged. At $t = 0$ a constant $+0.5$ V is applied. (a) Find the output slope in V/s. (b) How long until the output saturates? (c) Which direction does it ramp, and why?

Problem 8 (C) — Choosing a topology

A pH electrode behaves like a voltage source in series with $\sim 10^9$ Ω. You must amplify its signal by 5. (a) Inverting or noninverting topology — which is mandatory here, and why? (b) Bipolar-input or MOSFET-input op amp — which, and why? Support (b) with an estimate of the error a 50 nA bias current would cause.

Real Op-Amp Limitations

Problem 9 (A) — Offset budget

An inverting amplifier has $R_{\text{in}} = 2$ kΩ, $R_F = 100$ kΩ (gain -50). The op amp has $V_{OS} = 3$ mV, $I_B = 60$ nA, $I_{OS} = 10$ nA. (a) Compute the output error due to V_{OS} alone. (Careful: which gain multiplies V_{OS}?) (b) Add the bias-current error without a compensation resistor. (c) What compensation resistor should be added at V^+, and what does the total error become with it?

Problem 10 (A) — Gain–bandwidth budget

Your op amps have $f_T = 3$ MHz. A sensor signal occupies DC–15 kHz and needs a total gain of 200. (a) Show that a single stage fails the bandwidth requirement (compute its $f_{3\text{dB}}$ and the response at 15 kHz). (b) Show that two equal-gain stages succeed; include the composite-bandwidth shrinkage factor $\sqrt{\sqrt{2}-1} \approx 0.644$ for two identical first-order stages.

Problem 11 (A) — Slew rate

An op amp has $SR = 13$ V/μs. (a) What is the maximum frequency of an undistorted 5 V-peak sine output? (b) At 500 kHz, what is the largest undistorted peak amplitude? (c) A 100 mV-peak, 1 MHz sine: slew-limited or not (given $f_T = 10$ MHz)?

Op-Amp Varieties and Instrumentation Amplifiers

Problem 12 (C) — Programmable, then and now

The LM4250 is "programmed" with a resistor; a modern PGA (e.g., TI PGA113 or Microchip MCP6S91) is programmed over SPI. (a) What physical quantity does the LM4250's programming pin set, and name three specifications that scale with it. (b) What does the modern PGA's digital interface select instead? (c) Give one application where each style of programmability is the better fit.

Problem 13 (A) — Single-supply biasing

A single-supply op amp runs from +9 V and must buffer an audio signal of ±2.5 V riding on 0 V DC. (a) Why does connecting the signal directly fail? (b) Design a mid-rail bias using two equal 220 kΩ resistors and an input coupling capacitor; give the DC operating point and the output swing range. (c) Choose the coupling capacitor for a 20 Hz high-pass corner.

Problem 14 (A) — In-amp gain

For the standard three-op-amp instrumentation amplifier with first-stage resistors $R_1 = 25$ kΩ each and an output difference stage with $R_3/R_2 = 1$: (a) derive the expression for the overall differential gain; (b) find R_g for a gain of 500; (c) if both inputs rise together by 1 V (pure common mode), what does the first stage do with that volt, and why?

Problem 15 (C) — ECG front end

An ECG signal is ∼1 mV differential between two chest electrodes, each with 10–100 kΩ of variable skin-contact impedance, plus tens of millivolts of 60 Hz common-mode pickup. Explain *two distinct reasons* the one-op-amp difference amplifier is a poor front end here, and how the instrumentation amplifier's architecture fixes each one.

Filter Fundamentals

Problem 16 (A) — Bandpass bookkeeping

A bandpass filter has 3-dB frequencies $f_1 = 1.8$ kHz and $f_2 = 2.2$ kHz. Compute (a) the center frequency (geometric), (b) the bandwidth, (c) Q, and (d) the percent error of the arithmetic-mean estimate of f_0.

Problem 17 (A) — RC low-pass design

Design a first-order RC low-pass with $f_{3\text{dB}} = 500$ Hz using $C = 33$ nF. (a) Find R. (b) Compute the attenuation in dB at 5 kHz and at 10 kHz. (c) The load is 100 kΩ; qualitatively, what does connecting it do to the cutoff and the passband level, and which op-amp circuit removes the problem?

Problem 18 (C) — Cascading myth

"To get −40 dB/decade, just wire two identical RC low-pass sections in series." Explain what is right and what is wrong with this claim: what is the actual transfer function of the direct cascade, what happens to the corner, and what are *two* different fixes?

Problem 19 (C) — Passive vs. active

For each application, choose passive or active filtering and justify in one or two sentences: (a) a 100 MHz radio front-end preselector; (b) a 0.5 Hz high-pass to remove baseline drift from an ECG; (c) an audio crossover inside a passive (unpowered) loudspeaker cabinet.

Active Filter Design

Problem 20 (A) — Order selection

A low-pass filter must have $f_{3\text{dB}} = 2$ kHz and at least 45 dB of attenuation at 6 kHz. (a) Compute the required Butterworth order analytically. (b) What attenuation does the chosen order actually deliver at 6 kHz? (c) How many op amps will the realization need?

Problem 21 (A) — Scaling the table

Realize a third-order Butterworth low-pass with $f_{3\text{dB}} = 250$ Hz using the normalized three-pole section ($C_1 = 3.546$ F, $C_2 = 1.392$ F, $C_3 = 0.2024$ F, $R = 1\ \Omega$) and an impedance scale factor $Z = 2 \times 10^4$. Give all final component values.

Problem 22 (P) — Sallen–Key design and verification

Design a second-order Butterworth (Sallen–Key, unity gain, equal $R = 22$ kΩ) low-pass with $f_0 = 4$ kHz. (a) Use $\omega_0 = 1/(R\sqrt{C_1C_2})$ and $Q = \frac{1}{2}\sqrt{C_1/C_2} = 1/\sqrt{2}$ to find C_1 and C_2. (b) Write a Python script that plots the response and verifies −3 dB at 4 kHz and the attenuation one decade up. (c) What attenuation *should* appear at 40 kHz, from the rolloff rule alone?

Problem 23 (P) — Notch for mains interference

An ECG channel needs a 60 Hz notch with $Q = 8$. (a) Compute the notch bandwidth and the two 3-dB frequencies (use $f_{1,2} = f_0\left(\sqrt{1 + 1/4Q^2} \mp 1/2Q\right)$, and verify $f_2 - f_1 = f_0/Q$ and $\sqrt{f_1 f_2} = f_0$). (b) Write Python to plot $H(s) = (s^2 + \omega_0^2)/(s^2 + \frac{\omega_0}{Q}s + \omega_0^2)$ and confirm both 3-dB frequencies. (c) Why is a high Q desirable here — what would a $Q = 1$ notch do to the ECG?

Problem 24 (P) — Integrator simulation

An integrator with $R = 22\ \text{k}\Omega$ and $C = 4.7$ nF receives a ± 2 V square wave at 2 kHz. (a) Predict the output waveform shape and its peak-to-peak amplitude analytically. (b) Verify by numerical simulation (cumulative sum) and plot input and output. (c) What happens to the output if the input has a small DC offset, and what practical fix addresses it?

2 Solutions

Solution 1

With $A_o \approx 2 \times 10^5$ and ± 15 V supplies (saturation $\approx \pm 13.5$ V), the linear input window is only $|V^+ - V^-| < 13.5/2 \times 10^5 \approx 68\ \mu$V. A 1 mV signal exceeds this ~15-fold, so the output will not be an amplified copy: it slams between $+13.5$ V and -13.5 V every time the input crosses zero — a square wave carrying only the signal's sign, with all amplitude information destroyed. The one job a feedback-free op amp does well is exactly that: a **comparator** (which input is larger?).

Solution 2

(a) $|V_{\text{in}}| < 10.8/5 \times 10^4 = 216\ \mu$V, i.e., a window of $\pm 216\ \mu$V.
(b) $V_{\text{in}} = V_{\text{out}}/A_o = 5/5 \times 10^4 = 100\ \mu$V.
(c) The window is far smaller than realistic offsets, drift, and noise (V_{OS} alone is typically millivolts), so the operating point cannot be held inside it; without feedback there is no mechanism to stabilize the output at 5 V. Any practical amplifier must close a negative-feedback loop.

Solution 3

Rules: (1) the inputs draw no current; (2) with negative feedback, the output does whatever makes $V^+ = V^-$.
Invalid when: (i) **Positive feedback** (feedback to V^+, e.g., a Schmitt trigger): output reinforcement drives the amp into saturation instead of toward balance, so the inputs never equalize. (ii) **Saturated output** (demanded output beyond the rails, e.g., gain too high for the input): the output pins at V_{sat} and can no longer adjust to enforce the virtual short. (Also acceptable: slew-limited transients, or no feedback path at all — a comparator.)

Solution 4

(a) $R_{\text{in}} = 15$ kΩ (sets the input impedance), $R_F = 22 \times 15\text{k} = 330$ kΩ.
(b) Those *are* standard values: gain $= -330/15 = -22.0$ exactly; deviation 0% nominal (real deviation set by the 5% tolerances, worst case $\approx \pm 10\%$).
(c) $Z_{\text{in}} = R_{\text{in}} = 15$ kΩ: the inverting input is a virtual ground, so the source sees R_{in} straight to (virtual) ground. The noninverting topology's input connects to the op-amp input terminal itself (no resistor to ground in the signal path), hence its near-infinite input impedance.

Solution 5

(a) $G = 1 + 47/2.2 = 22.36$.
(b) $\beta = 1/22.36 = 0.04472$; $G_{\text{real}} = \dfrac{10^5}{1 + 10^5(0.04472)} = 22.355$. Error $= (22.364 - 22.355)/22.364 \approx 0.022\%$.
(c) $R_F = 0$ (and R_1 removed/open): a unity-gain **follower**, used as an impedance buffer — read a high-impedance node without loading it, drive a heavy load from a delicate source.

Solution 6

$V_{\text{out}} = -R_F(V_1/R_1 + V_2/R_2)$. Need $R_F/R_1 = 2 \Rightarrow R_1 = 50$ kΩ; $R_F/R_2 = 0.5 \Rightarrow R_2 = 200$ kΩ.
Check: $V_{\text{out}} = -(2 \times 1.5 + 0.5 \times (-2)) = -(3 - 1) = -2.0$ V.

Solution 7

(a) $|dV_{\text{out}}/dt| = V_{\text{in}}/RC = 0.5/(10^5 \times 10^{-7}) = 50$ V/s.
(b) $t = 11$ V/50 V s$^{-1} = 0.22$ s.
(c) Downward (toward -11 V): the positive input current flows into the virtual ground and out through the capacitor, forcing $V_{\text{out}} = -\frac{1}{RC}\int V_{\text{in}}\,dt$, negative for positive V_{in} — the integrator inverts.

Solution 8

(a) **Noninverting** (gain $1 + R_F/R_1 = 5$). The inverting topology's input impedance equals R_{in}; to avoid loading a 10^9 Ω source it would need $R_{\text{in}} \gg 10^9$ Ω, which makes bias-current and noise errors absurd. The noninverting input presents the op amp's own input impedance.
(b) **MOSFET (or JFET) input.** A bipolar bias current of 50 nA flowing from a 10^9 Ω source drops $I_B R_s = 50$ nA $\times 10^9$ Ω $= 50$ V (!) — complete nonsense; even 50 pA would drop 50 mV, swamping a pH electrode's ∼59 mV/pH. A MOSFET input at ∼0.1 pA drops ∼0.1 mV: acceptable.

Solution 9

(a) V_{OS} is amplified by the *noise gain* $1 + R_F/R_{\text{in}} = 51$ (not -50): 3 mV $\times$ 51 = 153 mV.
(b) Bias-current term $\approx I_B R_F = 60$ nA $\times$ 100 kΩ = 6 mV. Worst-case total ≈ 159 mV.
(c) $R_{\text{comp}} = R_{\text{in}} \parallel R_F = 2\text{k} \parallel 100\text{k} = 1.96$ kΩ from V^+ to ground. The current term drops to $I_{OS} R_F = 1$ mV; total ≈ 154 mV. Moral: here the offset *voltage* dominates; compensation fixes the current term only, so a lower-V_{OS} op amp (or trimming) is the real fix.

Solution 10

(a) Single stage: $f_{3\text{dB}} = f_T/G = 3 \times 10^6/200 = 15$ kHz — the band edge sits exactly at the -3 dB point, i.e., the top of the signal band is already attenuated 29% in voltage. Fails any flatness requirement.
(b) Two stages of gain $\sqrt{200} = 14.14$: each has $f_{3\text{dB}} = 3 \times 10^6/14.14 = 212$ kHz; composite bandwidth $\approx 0.644 \times 212 = 137$ kHz $\gg 15$ kHz. At 15 kHz each stage is down only $20\log_{10}\sqrt{1 + (15/212)^2} = 0.02$ dB; the pair, 0.04 dB. Passes easily.

Solution 11

(a) $f_{\max} = SR/(2\pi V_p) = 13 \times 10^6/(2\pi \times 5) = 414$ kHz.
(b) $V_p = SR/(2\pi f) = 13 \times 10^6/(2\pi \times 5 \times 10^5) = 4.14$ V.
(c) Required slew for 0.1 V at 1 MHz: $2\pi f V_p = 0.63$ V/μs $\ll 13$ V/μs — not slew-limited (and $f < f_T$, so not GBW-limited either at unity-ish gain). Small-signal speed and large-signal speed are separate budgets; always check both.

Solution 12

(a) The pin sets an internal quiescent *programming current* I_{set} (via an external resistor). Scaling with it: supply (quiescent) current/power, slew rate, gain–bandwidth, input bias current, noise (any three).
(b) Closed-loop *gain* (from an internal trimmed feedback network) and typically the *input channel* (built-in multiplexer); some parts add shutdown and calibration channels.
(c) LM4250 style: a battery instrument that must idle at microwatts and only needs modest speed (the resistor dials power down). PGA style: an auto-ranging data-acquisition front end where firmware must change gain on the fly between samples.

Solution 13

(a) The output (and the input common-mode range) cannot go below 0 V; the negative half of the ± 2.5 V swing would be clipped off at ground.
(b) Two 220 kΩ resistors from +9 V to ground bias V^+ at 4.5 V; the follower output sits at 4.5 V DC and swings 4.5 ± 2.5 V $= 2.0$–7.0 V, inside the rails.
(c) $R_{\text{eq}} = 220\text{k} \parallel 220\text{k} = 110$ kΩ; $C = 1/(2\pi f_c R_{\text{eq}}) = 1/(2\pi \times 20 \times 1.1 \times 10^5) = 72$ nF $\Rightarrow$ use 100 nF ($f_c = 14.5$ Hz).

Solution 14

(a) The virtual shorts place $V_{\text{in}+} - V_{\text{in}-}$ across R_g, so $i_g = (V_{\text{in}+} - V_{\text{in}-})/R_g$ flows through R_g and both R_1s (no current into op-amp inputs). Hence $V_{o2} - V_{o1} = i_g(2R_1 + R_g) = (V_{\text{in}+} - V_{\text{in}-})(1 + 2R_1/R_g)$, and the unity difference stage passes it: $G = (1 + 2R_1/R_g) \cdot 1$.
(b) $500 = 1 + 50\text{k}/R_g \Rightarrow R_g = 50\,000/499 = 100.2\ \Omega$ (use 100 Ω; $G = 501$).
(c) Both buffer outputs rise by exactly 1 V (R_g carries no current since both ends move together): the first stage passes common mode at *unity* gain while amplifying the difference by 500 — that asymmetry is the in-amp's whole trick, leaving the difference stage a 500-times-easier rejection job.

Solution 15

Reason 1 — input loading. The difference amp's inputs look into its input resistors (R_1); with 10–100 kΩ of *variable, unequal* electrode impedance in series, the effective resistor ratios change with skin contact, corrupting both the gain and the balance. *Fix:* the in-amp's two input op amps present near-infinite input impedance, so electrode impedance drops (almost) no voltage.
Reason 2 — common-mode rejection by resistor matching. The single-amp stage rejects the 60 Hz common mode only as well as its four resistors match; 1% resistors give only $\sim$40 dB. *Fix:* the in-amp's first stage amplifies the differential signal (e.g., $\times 500$) while passing common mode at unity, so the final stage's finite matching acts on a signal that is already hundreds of times larger than the interference; integrated, laser-trimmed resistors push CMRR past 100 dB.

Solution 16

(a) $f_0 = \sqrt{1800 \times 2200} = 1990$ Hz. (b) $BW = 400$ Hz. (c) $Q = 1990/400 = 4.97$. (d) Arithmetic mean $= 2000$ Hz; error $= (2000 - 1990)/1990 = 0.5\%$ — tiny here because $f_2/f_1 = 1.22$ is a fairly narrow band.

Solution 17

(a) $R = 1/(2\pi f_c C) = 1/(2\pi \times 500 \times 33 \times 10^{-9}) = 9.65$ kΩ (use 9.53 k or 10 k).
(b) At $10f_c$: $|H| = 1/\sqrt{1+100} \Rightarrow -20.04$ dB. At $20f_c$: $1/\sqrt{1+400} \Rightarrow -26.0$ dB.
(c) The 100 kΩ load appears in parallel with C (and, at DC, with nothing else to ground), forming a divider with R: the passband level drops (to $\approx 100/(100+9.65) = 0.91$, i.e., -0.8 dB) and the corner shifts upward (Thévenin resistance $R \parallel R_L < R$). A **voltage follower** between filter and load removes both effects.

Solution 18

Right: far above the corner the asymptotic slope is indeed -40 dB/decade.
Wrong: the second section loads the first, so the cascade is *not* $1/(1+sRC)^2$; nodal analysis gives $H(s) = 1/\big(1 + 3sRC + (sRC)^2\big)$ — the "3" (instead of 2) splits the poles apart, drags the composite -3 dB corner well below $1/2\pi RC$, and rounds the knee.
Fixes: (1) buffer between sections with a voltage follower (restores the ideal product); (2) make the second section's impedance much larger than the first's (e.g., $10R$, $C/10$) so loading is negligible; or best, (3) use an active (Sallen–Key) second-order section with the proper Butterworth Q, which is flatter than *any* buffered equal-RC pair.

Solution 19

(a) **Passive** (LC): 100 MHz is far above op-amp GBW territory; passive RF filters are standard, small, and noiseless.
(b) **Active**: a 0.5 Hz passive corner needs enormous components ($RC = 0.32$ s), and the following stage would load it; an op-amp high-pass (or an active integrator-feedback topology) achieves it with practical R and C and a buffered output.
(c) **Passive**: there is no power supply inside an unpowered cabinet, and the filter must pass loudspeaker-level power — an inductor/capacitor crossover is the only option.

Solution 20

(a) $A_s^{(\omega)} = 6/2 = 3$; $n \geq \dfrac{\log_{10}(10^{4.5} - 1)}{2\log_{10} 3} = \dfrac{4.4999}{0.9542} = 4.72 \Rightarrow n = 5$.
(b) $-10\log_{10}(1 + 3^{10}) = -10\log_{10}(59050) = -47.7$ dB.
(c) An $n = 5$ realization is a three-pole section plus a two-pole section: **2 op amps**.

Solution 21

Scale factor for capacitors: $Z \cdot 2\pi f_{3\text{dB}} = 2 \times 10^4 \times 2\pi \times 250 = 3.1416 \times 10^7$.

$$C_1 = \frac{3.546}{3.1416 \times 10^7} = 112.9\,\text{nF}, \quad C_2 = \frac{1.392}{3.1416 \times 10^7} = 44.3\,\text{nF}, \quad C_3 = \frac{0.2024}{3.1416 \times 10^7} = 6.44\,\text{nF},$$

and all three resistors become $R = Z \times 1\ \Omega = 20$ kΩ. (Practical picks: 110 nF, 44 nF (2×22 nF), 6.8 nF, 20 kΩ.)

Solution 22

(a) $Q = 1/\sqrt{2} \Rightarrow C_1/C_2 = 4Q^2 = 2$; $\sqrt{C_1 C_2} = 1/(\omega_0 R) = 1/(2\pi \times 4000 \times 22\,000) = 1.809$ nF. So $C_1 = \sqrt{2} \times 1.809 = 2.56$ nF, $C_2 = 1.809/\sqrt{2} = 1.28$ nF (practical: 2.7 nF and 1.2 nF).

(b) Script and output plot (Figure 1): the trace shows -3.01 dB at 4 kHz.

```python
import numpy as np, matplotlib.pyplot as plt
from scipy import signal
R, C1, C2 = 22e3, 2.558e-9, 1.279e-9
w0 = 1/(R*np.sqrt(C1*C2)); Q = 0.5*np.sqrt(C1/C2)
b, a = [w0**2], [1, w0/Q, w0**2]
f = np.logspace(2, 5.5, 600)
_, h = signal.freqs(b, a, 2*np.pi*f)
for ff in (4e3, 40e3):
    _, hh = signal.freqs(b, a, [2*np.pi*ff])
    print(f"{ff:8.0f} Hz: {20*np.log10(abs(hh[0])):7.2f} dB")
plt.semilogx(f, 20*np.log10(abs(h))); plt.grid(True); plt.show()
#    4000 Hz:   -3.01 dB
#   40000 Hz:  -40.01 dB
```

(c) Second order $\Rightarrow -40$ dB/decade; one decade above the corner should sit ≈ 40 dB down — matching the simulated -40.0 dB.

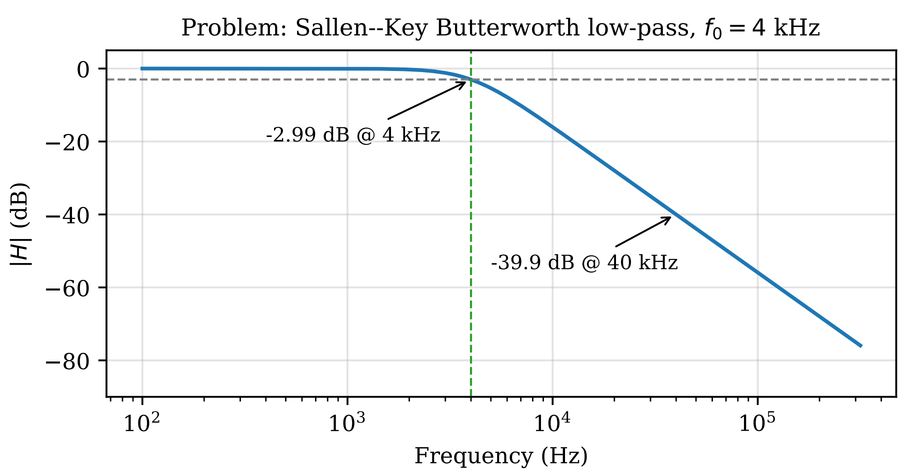

Figure 1: Solution 22: simulated response of the 4 kHz Sallen–Key design.

Solution 23

(a) $BW = f_0/Q = 60/8 = 7.5$ Hz. With $1/2Q = 0.0625$ and $\sqrt{1 + 1/4Q^2} = 1.00195$: $f_1 = 60(1.00195 - 0.0625) = 56.37$ Hz, $f_2 = 60(1.00195 + 0.0625) = 63.87$ Hz. Checks: $f_2 - f_1 = 7.50$ Hz ✓; $\sqrt{f_1 f_2} = \sqrt{56.37 \times 63.87} = 60.00$ Hz ✓.

(b) See the script below and Figure 2; the simulated -3 dB points land on 56.4 and 63.9 Hz.

```python
import numpy as np, matplotlib.pyplot as plt
from scipy import signal
f0, Q = 60.0, 8.0
w0 = 2*np.pi*f0
b, a = [1, 0, w0**2], [1, w0/Q, w0**2]
f = np.linspace(10, 200, 5000)
_, h = signal.freqs(b, a, 2*np.pi*f)
mag = 20*np.log10(np.maximum(abs(h), 1e-12))
lo = f[f < 60][np.argmin(abs(mag[f < 60] + 3))]
hi = f[f > 60][np.argmin(abs(mag[f > 60] + 3))]
print(lo, hi)          # 56.4  63.9  Hz
plt.plot(f, mag); plt.grid(True); plt.show()
```

(c) The ECG's diagnostic content spans roughly 0.5–100 Hz, straddling 60 Hz. A $Q = 1$ notch ($BW = 60$ Hz, roughly 35–103 Hz at -3 dB) would gouge out a huge slice of the signal band and distort the QRS complex; $Q = 8$ removes only a 7.5 Hz sliver around the interference.

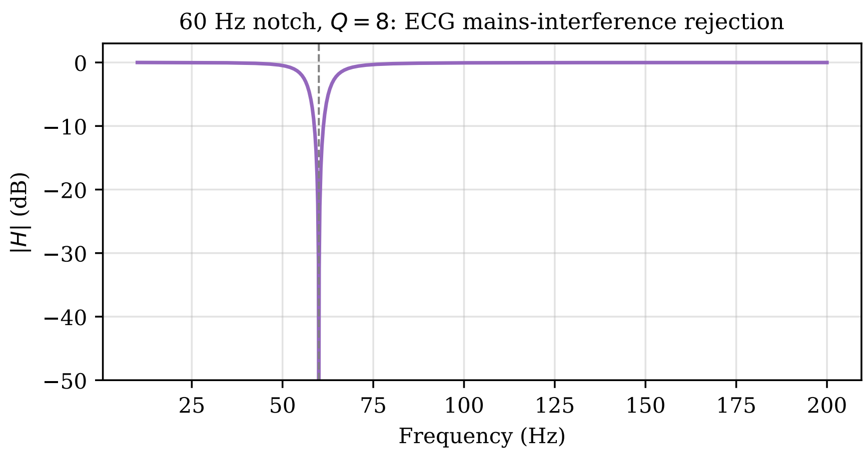

Figure 2: Solution 23: 60 Hz, $Q = 8$ notch response.

Solution 24

(a) During each half-period the input is constant at ± 2 V, so the output ramps at $|V_{\text{in}}|/RC = 2/(22\,000 \times 4.7 \times 10^{-9}) = 1.934 \times 10^4$ V/s. Over a half-period $T/2 = 0.25$ ms the ramp travels $1.934 \times 10^4 \times 2.5 \times 10^{-4} = 4.84$ V: a **triangle wave of 4.84 V peak-to-peak** (inverted phase: input high $\Rightarrow$ output ramping down).

(b) Simulation (Figure 3) reproduces 4.85 V pk–pk:

```python
import numpy as np, matplotlib.pyplot as plt
from scipy import signal
fs = 2e6
t  = np.arange(0, 2e-3, 1/fs)
vin  = 2*signal.square(2*np.pi*2e3*t)
R, C = 22e3, 4.7e-9
```

```python
vout = -np.cumsum(vin)/fs/(R*C)
print(vout.max() - vout.min())    # ~4.85 V
plt.plot(t*1e3, vin); plt.plot(t*1e3, vout); plt.show()
```

(c) Any DC offset integrates into a steady ramp that eventually saturates the op amp (the capacitor also integrates bias current). Practical fix: a large "bleed" resistor in parallel with C (turning the circuit into a low-frequency-limited integrator), or periodic reset of the capacitor.

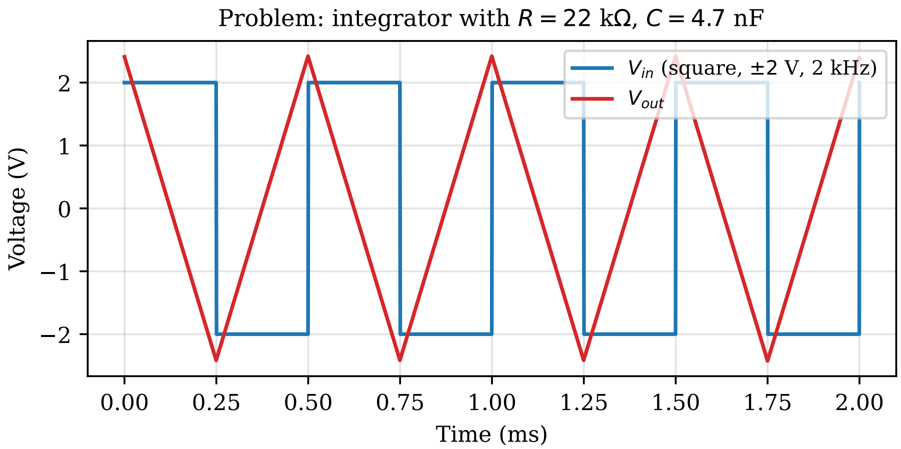

Figure 3: Solution 24: integrator input and output.